

Risk Methods — Quick Reference Card

VaR · ES · Backtests · Sensitivities · P&L attribution · Estimation · Stress · Capital. Deep companion: risk\_methods\_cheatsheet.pdf

◆ Universal Symbols

|            |                                          |
|------------|------------------------------------------|
| α          | confidence (0.99 = 99%); tail = 1−α      |
| s          | sensitivity vector (∂V/∂f_i)             |
| Σ          | factor cov matrix (units must match s)   |
| F          | history matrix (rows=days, cols=factors) |
| z(α)       | norm. ppf(α); 99%=2.326, 99.5%=2.576     |
| N(·), ϕ(·) | std-normal CDF, PDF                      |

Sign: loss is POSITIVE (VaR, ES). P&L negative when you lose.

◆ Master Recipe (4 steps)

1. Identify factors — rate, spread, vol, FX, equity (bucket KRD per tenor)

2. Generate shocks — history / parametric (z·σ) / MC (Cholesky)

3. Revalue — sensitivity (linear) OR full-reval (exact)

4. Aggregate — quantile (VaR) / mean-of-tail (ES) / gradient (MVar)

METHOD PICKER

| Method         | Shocks       | Reval      |
|----------------|--------------|------------|
| Historical VaR | F (replay)   | FULL       |
| Parametric     | Σ implied    | sens (lin) |
| Monte Carlo    | chol(Σ)·z    | FULL       |
| Filtered Hist  | F·σ_t/σ_then | FULL       |
| Cornish-Fisher | Σ + S, K     | sens (lin) |

◆ Three patterns

P1 — SENSITIVITY-BASED

risk ≈ s·shock or √(s<sup>T</sup>Σs)

Fast O(#factors); falls down on non-linearity.

P2 — FULL REVALUATION

risk = T( V(shocked) − V(base) over many shocks )

Exact; cost = n\_paths × pricer cost.

P3 — EMPIRICAL DISTRIBUTION

risk = T( empirical P&L vector )

No distribution assumption; bounded by window.

Historical = P2+P3; Parametric = P1; MC = P2+simulated; FHS = P2+P3+GARCH.

◆ Reference numbers (drill 13 portfolio)

| Metric                            | Value                           |
|-----------------------------------|---------------------------------|
| P&L std σ = √(s <sup>T</sup> Σs)  | 2.5327                          |
| Historical VaR(99.5%)             | 4.6781                          |
| Parametric VaR(99.5% / 99%)       | 6.5238 / 5.8919                 |
| MC VaR(99.5% / 99%) seed=42       | 6.3411 / 5.8760                 |
| Gammas (rate, spread, vol, fx)    | 1.86e-4, 4.29e-5, −1.06, 412.92 |
| 5y bond hist VaR(99.5%) [drill 8] | 0.7363                          |

4.68 (hist) < 6.34 (MC) < 6.52 (param): hist's bounded tail is THINNER than Gaussian; MC's full-reval captures gamma that param's linear misses.

● I — VaR & Expected Shortfall

Historical VaR (§1)

VaR(α) = −sorted(P&L)[[(1−α)·N]]

P3 — empirical. No distribution assumption.

np.percentile(pnl, (1-conf)\*100) \* -1

Limits: bounded by window; vol-blind; calm year → tiny VaR.

Parametric / delta-normal VaR (§2)

σ<sup>2</sup> = s<sup>T</sup> Σ s

VaR(α) = z(α) · σ

P1 — analytical Normal. Fast.

norm.ppf(conf) \* np.sqrt(s @ cov @ s)

Assumes: factors Normal + book LINEAR. Misses fat tails + gamma.

Monte Carlo VaR (§3)

L = chol(Σ); shocks = z·L<sup>T</sup> ~ N(0,Σ)

P&L k = V(base + shock k) − V<sub>0</sub> (FULL reval)

VaR(α) = −percentile(P&L, (1−α)·100)

P2 + simulated. Captures gamma + custom distributions.

L = np.linalg.cholesky(cov)  
z = rng.standard\_normal((n,4))  
shocks = z @ L.T

Always set seed for reproducible risk numbers.

Expected Shortfall (CVaR) (§4)

Empirical: ES α = −mean(sorted[:,(1−α)N])

Normal: ES α = σ · ϕ(z(α)) / (1−α)

FRTB: ES at α=0.975

Coherent (subadditive). Mean of tail, not boundary.

k = int(np.ceil((1-conf)\*len(pnl)))

np.sort(pnl)[:k].mean()

ES/VaR (Normal): α=0.99 → 1.145x; α=0.975 → 1.193x.

Filtered Historical Sim (FHS) (§5)

shock\_today = (σ\_today / σ\_then) · F\_i

then proceed as historical VaR

P2+P3+GARCH rescale. Solves "stale during crises".

Pitfall: GARCH mis-spec → wrong rescale. Plot ô\_t.

Cornish-Fisher VaR (§6)

z\_CF = z + (z<sup>2</sup>−1)S/6 + (z<sup>3</sup>−3z)K/24 − (2z<sup>3</sup>−5z)S<sup>2</sup>/36

VaR = −(μ + z CF·σ)

Quick fat-tail fix without MC. S=skew, K=excess kurt.

Falls apart for |S|>2 or K>10 — use EVT/t-copula.

Marginal / Component / Incremental VaR (§7)

MVaR\_i = z(α)·(Σw)\_i / √(w<sup>T</sup>Σw)

CVaR\_i = w\_i · MVaR\_i [Σ CVaR\_i = VaR ✓ Euler]

IVaR\_i = VaR(w) − VaR(w\_{−i})

M = per-\$ trade sizing; C = limit attribution; I = "if I close X".

I Σ here = POSITION-return cov (N×N), NOT factor cov. Σ\_pos = B·Σ\_factor·B<sup>T</sup>.

Risk Budgeting / ERC (§7b)

solve w : w\_i·(Σw)\_i / (w<sup>T</sup>Σw) = b\_i ∀i

ERC special case: b\_i = 1/N

Maillard iteration: w\_i ← target/(w\_i·(Σw)\_i); renormalise

Inverse of §7. SAA when expected returns uncertain. Risk-parity funds (Bridgewater).

w\_new = w \* (target / (w \* (cov @ w)))

w\_new /= w\_new.sum()

Stressed VaR (sVaR) (§8)

Historical VaR on FIXED 12M stress window

window chosen for MAX sVaR on CURRENT book

Basel II.5: Capital = max(VaR\_t, k·avg<sub>60</sub>) + max(sVaR\_t, k·avg<sub>60</sub> sVaR)

Example: calm 250d → VaR 2.3; 2008 window → sVaR 9.2 → capital ~5x higher.

▲ II — Backtesting

Kupiec POF (§9)

LR = 2[x·ln(p/π) + (T−x)·ln((1−p)/(1−π))]

~ χ<sup>2</sup>(1); reject if LR > 3.841

π = 1−α, p = x/T, x = #hits

Unconditional hit-rate test. Ignores clustering.

Christoffersen independence (§10)

LR\_ind ~ χ<sup>2</sup>(1) on 2x2 transition table

LR\_CC = LR POF + LR\_ind ~ χ<sup>2</sup>(2)

Catches vol-blind model: 6 hits OK count but all in 1 week.

Basel traffic light (§11)

| Hits | Zone   | Multiplier |
|------|--------|------------|
| 0–4  | GREEN  | 3.00       |
| 5    | YELLOW | 3.40       |
| 6    | YELLOW | 3.50       |
| 7    | YELLOW | 3.65       |
| 8    | YELLOW | 3.75       |
| 9    | YELLOW | 3.85       |
| ≥10  | RED    | 4.00       |

99% 1-day VaR over 250d. Capital = avg\_60d · k.

ES backtest (Acerbi-Szekely T2) (§12)

Z = (1/T)·Σ[X·I/(ES·(1−α))] − 1

H<sub>0</sub>: E[Z]=0; Z < 0 (negative) = under-est

Mandated by FRTB IMA. Low power at α=0.975 (~6 hits / 250d).

■ III — Risk Sensitivities

First-order (DV01, CS01, vega, FX-Δ, IE01) (§13)

s\_f = [V(base + δ\_f) − V(base)] / δ\_f

ΔV ≈ Σ f s\_f · shock\_f

| Greek | Factor              | Bump             |
|-------|---------------------|------------------|
| DV01  | parallel rate       | +1 bp            |
| CS01  | credit spread       | +1 bp            |
| IE01  | breakeven inflation | +1 bp            |
| FX-Δ  | FX shock            | +1% (0.01)       |
| Vega  | implied vol         | +1 vol pt (0.01) |

Convention: "DV01 = V\_bumped − V\_base" → long bond DV01 is NEGATIVE. The other convention (loss-given-up) flips sign. Pick one and LABEL it.

Second-order (gamma, convexity, cross-gamma) (§14)

g\_f = [V(+δ) − 2V<sub>0</sub> + V(−δ)] / δ<sup>2</sup> own

g\_{ab} = [V(+a+b) − V(+a−b) − V(−a+b) + V(−a−b)] / (4·δ\_a·δ\_b) cross

Drill 12 gammas: (1.86e-4, 4.29e-5, −1.06, 412.92). Adding 2nd-order shrank residual 4x.

Key Rate Duration (KRD) (§15)

KRD k = V(bump z(T k) +1bp) − V(base)

Σ KRD\_k ≈ parallel DV01

Curve-shape risk. Hedge gamma, not scalar.

Pension fail mode: 5y IRS hedging 30y liab = parallel-neutral, drowns in steepener.

Computation methods (§16)

| Method        | Cost                             |
|---------------|----------------------------------|
| Forward FD    | 1·n_factors                      |
| Central FD    | 2·n_factors (O(δ <sup>2</sup> )) |
| AAD (adjoint) | ~3–5x 1 fwd pass for ALL         |

FRTB SBM with 1000s of buckets → AAD pays off above ~50 factors.

Black-76 Greeks (forward F) (§17)

|   |                             |
|---|-----------------------------|
| Δ | DF·N(d1)                    |
| Γ | DF·ϕ(d1) / (F·σ·√T)         |
| v | DF·F·ϕ(d1)·√T               |
| Θ | − for long opt (time decay) |

Γ uses F not S in denominator. Free from price formula.

◆ IV — P&L Attribution

Greek-based Taylor explain (§18)

first\_f = s\_f · Δf

second\_f = ½ · g\_f · Δf<sup>2</sup>

unexp = full − Σfirst − Σsecond

Drill 12 result: full −3.024 / first −3.054 / 2nd +0.023 / unexp +0.008.

Rule of thumb: |unexp| < 5% |full| for vanilla derivs.

Risk-factor explain (RFE) (§19)

Sens RFE: contrib\_f = s\_f · Δf (order-indep, misses cross)

Full RFE: contrib\_f = V(only Δf) − V(base) (+ residual)

Sequential: apply IN ORDER; Σ = full, order-dep

Vanilla → sens. Options → full + cross-gamma report. Regulator → sequential.

Actual / Hyp / RT P&L (§20)

Actual = V\_eod(pos\_eod, mkt\_eod) − V\_sod(pos\_sod, mkt\_sod)

Hyp = V(pos\_sod, mkt\_eod) − V(pos\_sod, mkt\_sod)

RT (FRTB) = V\_VaR\_pricer(pos\_sod, ...)

Unexp\_intraday = Actual − Hyp

PLA-residual = Hyp − RT

Two residuals: desk behaviour (unexp\_intraday) vs model quality (PLA-residual).

FRTB PLA test (§21)

p\_s ≥ 0.80 / 0.70 / <0.70 = G/A/R

K5 < 0.09 / 0.12 / ≥0.12 = G/A/R

Joint zone = WORST of the two

RED → desk excluded from IMA → SA capital fallback.

spearmanr(hyp, rt); ks\_2samp(hyp, rt)

Carry / pull-to-par / theta (§22)

Time\_PnL = V(mkt\_today, t+1) − V(mkt\_today, t)

= carry + pull\_to\_par + theta + cross-roll

"What would have happened with NO market move." Carry is EXPECTED daily P&L.

● V — Estimation Inputs

Historical vol (§23)

$\sigma^2 = (1/(N-1)) \sum (r - \bar{r})^2$   
 $\sigma^{\text{annual}} = \sigma^{\text{daily}} \cdot \sqrt{252}$

Equal weights. Stale during regime shifts.

EWMA (RiskMetrics) (§24)

$\sigma^2_t = \lambda \cdot \sigma^2_{t-1} + (1-\lambda) \cdot r^2_{t-1}$   
 $\lambda = 0.94$  daily, 0.97 monthly  
half-life =  $\ln(0.5)/\ln(\lambda) = 11.2d$  @ 0.94

One parameter; vol updates immediately. No mean reversion.

GARCH(1,1) (§25)

$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1}$   
 $\sigma^2_\infty = \omega/(1-\alpha-\beta)$ ,  $\alpha+\beta < 1$  stationary  
 $\alpha+\beta = 1 \equiv \text{EWMA (IGARCH)}$

Adds mean reversion. Forecast  $\hat{\sigma}_{t+h} \rightarrow \sigma_\infty$ .  
 $\text{arch\_model}(r*100, \text{vol}=\text{"GARCH"}, p=1, q=1).\text{fit}()$

Implied vol (§26)

Solve  $\sigma$ :  $\text{BS}(\sigma) = \text{market\_price}$   
Brent on  $f(\sigma)=\text{BS}(\sigma)-\text{mkt}$ ;  $\sigma \in [1e-6, 5]$

Forward-looking; surface = vol by strike  $\times$  expiry. Use Brent over Newton at boundaries.

Sample covariance (§27)

$\Sigma^{\wedge} = \text{cov}(R, \text{rowvar=False}, \text{ddof}=1)$

Gold standard ONLY when  $T \gg N$ .  $T < N \rightarrow$  singular.

Ledoit-Wolf shrinkage (§28)

$\Sigma^{\wedge}\text{shrink} = \delta \cdot F + (1-\delta) \cdot S$   
 $\delta^* = \max(0, \min(1, \pi/\gamma^2))$

Regularises  $\Sigma$  when  $N \approx T$ .  $\delta \approx 0.1-0.4$  for daily equity.

$\text{LedoitWolf}().\text{fit}(R).\text{covariance\_}$

PSD repair (§29)

$\Sigma = Q\Lambda Q^T$ ; clip  $\lambda_i$  to  $\varepsilon$   
 $\Sigma_{\text{psd}} = Q \cdot \text{diag}(\max(\lambda, \varepsilon)) \cdot Q^T$   
rescale diagonal to preserve variances

Standard fix when cov is "almost-PSD" (numerical noise, missing data, bumps).

▲ VI — Stress & Scenarios

Historical scenario library (§30)

| Event            | Headline shocks                   |
|------------------|-----------------------------------|
| Black Monday '87 | S&P −22%, vol $\times 3$          |
| LTCM '98         | EMBI +800bp, USTs flight          |
| Lehman week '08  | Libor-OIS +200, S&P −12, USTs −80 |
| COVID Mar '20    | S&P −34, VIX 80, IG +250, HY +600 |
| Gilt LDI '22     | 30y +120bp/3d, GBP −5             |

Hypothetical templates (§31)

Parallel  $\pm 100\text{bp} \cdot 2s10s$  steepener/flattener  
IG +200bp / HY +500bp blowout  
vol +50% relative  $\cdot$  FX  $\pm 10\%$  ( $3\sigma$ )  
risk-off combo: rate −50, spread +100, vol +30%

Solvency II SCR uses prescribed 1-in-200y deterministic shocks.

Reverse stress test (§32)

find  $s^*$  minimising  $s^T \Sigma^{-1} s$  subject to  $\text{PnL}(s) = -L^*$   
Linear closed-form:  $s^* = (-L^*/s^T \Sigma s) \cdot \Sigma \cdot \text{sens}$   
Mahalanobis =  $L^* / \sigma_{\text{pnl}}$

"Find cheapest way to lose  $\$X$ ". Hedged book = high Mahalanobis. ICAAP requirement.

Aggregating scenarios (§33)

Report: worst case ; full table  
DON'T: average / sum

Why no sum: Lehman + COVID both contain equity shock  $\rightarrow$  adding implies it happens TWICE. Overstates loss 50-200%.

◆ VII — Capital Frameworks

Basel SA-CCR (§34)

EAD =  $1.4 \cdot (\text{RC} + \text{PFE})$   
 $\text{RC} = \max(V-C, 0)$   
 $\text{PFE} = \text{mult} \cdot \sum \text{AddOn\_AC}$   
 $\text{mult} = \min(1, 0.05 + 0.95 \cdot \exp((V-C)/(2 \cdot \sum \text{AddOn})))$

Replaces CEM. Counterparty exposure for OTC + SFTs.

Basel IMA legacy (§35)

$\text{MRC} = \max(\text{VaR}_t, k \cdot \text{avg}_{60s} \text{VaR})$   
 $\quad + \max(s\text{VaR}_t, k \cdot \text{avg}_{60s} \text{VaR})$   
 $k \in [3.0, 4.0]$  (traffic light)

Phased out by FRTB Jan 2025. Still used in some emerging-market regulators.

FRTB SA (SBM) (§36)

Delta:  $\sqrt{(\sum \text{WS}_b^2 + \sum \{b \neq c\} \rho_{bc} \cdot \text{WS}_b \cdot \text{WS}_c)}$   
Vega: same on vega sens  
Curvature:  $\max(\text{CVR}_{\text{up}}, \text{CVR}_{\text{down}})^2$  aggregated  
 $+ \text{DRC} + \text{RRA0}$

Sensitivity-tree: 5 risk classes  $\times$  dozens of buckets  $\times$  per-bucket aggregation.

FRTB IMA (§37)

IMCC =  $0.5 \cdot \text{ES}_{\text{full stressed}} + 0.5 \cdot \text{ES}_{\text{reduced}}$   
ES at  $\alpha=0.975$  with LH scaling  
IMA = IMCC + SES (NMRF) + DRC

$\text{ES}(97.5) \approx \text{VaR}(99)$  under Normal. Desk-level PLA + backtest required.

Operational cost:

most banks dropped IMA for many desks; SA fallback.

IRC / DRC (default risk) (§38)

IRC: MC on Vasicek ASRF, 99.9%/1y  
DRC:  $\text{JTD}_i = \text{LGD} \cdot \text{notional} - \text{MtM}$   
 $\text{DRC} = \sum \text{buckets} \max(\Sigma \text{longs}, |\Sigma \text{shorts}|)$

Default = jump, missed by VaR. Concentrated single-names get hammered.

◆ Coherence axioms (Artzner)

| # | Axiom                  | VaR | ES |
|---|------------------------|-----|----|
| 1 | Monotonicity           | ✓   | ✓  |
| 2 | Translation invariance | ✓   | ✓  |
| 3 | Positive homogeneity   | ✓   | ✓  |
| 4 | Subadditivity          | ✗   | ✓  |

VaR can punish diversification.

Two indep low-prob big-loss assets: indiv  
 $\text{VaR}(0.95)=0$ , joint  $\text{VaR}(0.95)>0$ . ES never. **FRTB chose ES for this reason.**

**Use:** ES for capital, VaR for limits.

◆ Risk-factor sign cheatsheet

| Long position                | DV01            | CS01 | IE01 | FX-Δ | Vega |
|------------------------------|-----------------|------|------|------|------|
| Vanilla bond                 | −               | −    | 0    | 0    | 0    |
| Linker                       | −               | 0    | +    | 0    | 0    |
| Foreign bond                 | −               | −    | 0    | +    | 0    |
| FRN (between resets)         | $\approx 0$     | −    | 0    | 0    | 0    |
| IRS receive-fixed            | −               | 0    | 0    | 0    | 0    |
| IRS pay-fixed                | +               | 0    | 0    | 0    | 0    |
| ZCIS receive-infl            | $\approx 0$ par | 0    | +    | 0    | 0    |
| FX Spot/Fwd (long for)       | + small         | 0    | 0    | +    | 0    |
| CDS protection BUYER         | − small         | +    | 0    | 0    | 0    |
| CDS protection SELLER        | + small         | −    | 0    | 0    | 0    |
| Long call (FX/swaption)      | small           | 0    | 0    | +Δ   | +    |
| Long put                     | small           | 0    | 0    | −Δ   | +    |
| Pension annuity liab (short) | +               | 0    | −    | 0    | 0    |

Vendor convention:

Bloomberg / risk-systems DV01 is LOSS-given-up (long bond = +). FLIP every sign in DV01 column above before plugging into  $s \cdot \Delta r$ .

◆ Top 6 traps (6-month memory)

Units must match

between  $s$  and  $\Sigma$  (bp vs decimal vs %). Always check  $\sigma_{\text{pnl}}$  in book units.

DV01 sign convention

— pick  $\partial V/\partial r$  or loss-given-up; label EVERY report.

250-day window

at  $\alpha=0.995 \rightarrow \text{VaR} = 2\text{nd worst day} \rightarrow$  enormous sampling error.

Gamma uses F (forward), not S (spot)

in Black-76 denominator.

Always set MC seed

— risk numbers must be reproducible day-to-day.

VaR not subadditive

$\rightarrow$  can punish diversification on heavy-tail / discrete-loss books. Use ES for capital.

2 / 2